

Finite generation of canonical rings II

Christopher Hacon

University of Utah

December, 2006

Outline of the talk

1 The structure of the proof

Outline of the talk

- 1 The structure of the proof
- 2 The directed MMP

Outline of the talk

- 1 The structure of the proof
- 2 The directed MMP
- 3 Existence of models

Outline of the talk

- 1 The structure of the proof
- 2 The directed MMP
- 3 Existence of models
- 4 Finiteness of models

Outline of the talk

- 1 The structure of the proof
- 2 The directed MMP
- 3 Existence of models
- 4 Finiteness of models
- 5 Non-vanishing

Outline of the talk

- 1 The structure of the proof
 - The structure of the proof
- 2 The directed MMP
- 3 Existence of models
- 4 Finiteness of models
- 5 Non-vanishing

The rough structure of the proof

- Recall that there are 3 main theorems we have to prove:
- Thm A. Existence of log terminal models
- Thm B. Nonvanishing
- Thm C. Finiteness of models
- Roughly speaking we would like to show that:
 - $A_{n-1} + B_{n-1} + C_{n-1}$ imply A_n ;
 - $B_{n-1} + C_{n-1} + A_n$ imply B_n ;
 - $C_{n-1} + A_n + B_n$ imply C_n .
- We will use the fact that by a result of Hacon-M^cKernan,
 $A_{n-1} + B_{n-1} + C_{n-1}$ imply that PL-flips exist in dimension n .

The structure of the proof I

- We now recall the precise statements of Thm's A, B, C:
- Let $\pi : X \rightarrow U$ be a projective morphism of normal quasi-projective varieties, where $\dim(X) = n$.
- **Thm A.** (Existence of log terminal models.) *Suppose that $K_X + \Delta$ is KLT and Δ is big over U .
 If $K_X + \Delta \sim_{\mathbb{R}, U} D \geq 0$, then $K_X + \Delta$ has a log terminal model over U .*
- **Thm B.** (Non-vanishing.) *Suppose that $K_X + \Delta$ is KLT and Δ is big over U .
 If $K_X + \Delta$ is π -pseudo-effective over U , then there is an effective $\mathbb{R}$ -divisor D such that $K_X + \Delta \sim_{\mathbb{R}, U} D$.*

The structure of the proof II

- **Thm C. (resp. C^-)** (Finiteness of models.) *Fix A an ample divisor over U . Suppose that there exists a divisor Δ_0 such that $K_X + \Delta_0$ is KLT.*
Let $\mathcal{C} \subset \{\Delta \mid K_X + \Delta + A \text{ is log canonical}\}$ be a rational polytope (resp. a rational polytope contained in the big cone).
Then the set of isomorphism classes

$$\{Y \mid Y \text{ is a WLCM for } (X, \Delta + A) \text{ over } U; \Delta + A \in \mathcal{C}\}$$

is finite.

- **Claim 1.** Theorems A_{n-1} , B_{n-1} and C_{n-1} imply Theorem A_n
- **Claim 2.** Theorem A_n and Theorem C_{n-1} imply Theorem C_n^- ,
- **Claim 3.** Theorem A_n and Theorem C_n^- imply Theorem B_n ,
- **Claim 4.** Theorem A_n and Theorem B_n imply Theorem C_n .

Remarks

- This is joint work with C. Birkar, P. Cascini and J. McKernan.
- We work over the field of complex numbers $\mathbb{C}$.
- Throughout the proof I will usually assume for simplicity that $U = \text{Spec} \mathbb{C}$.

Outline of the talk

- 1 The structure of the proof
- 2 The directed MMP
- 3 Existence of models
- 4 Finiteness of models
- 5 Non-vanishing

The directed MMP

- The main idea of the proof of Claim 1 is to show that any sequence of flips with scaling terminate.
- To run the MMP with scaling, assume that X is $\mathbb{Q}$ -factorial, (X, Δ) is KLT and $C \geq 0$ is an $\mathbb{R}$ -divisor such that $K_X + \Delta + C$ is nef and KLT.
- Let $\lambda = \inf\{t \geq 0 \mid K_X + \Delta + tC\} \text{ is nef}$
- If $\lambda = 0$ we STOP. Otherwise there is an extremal ray R such that $(K_X + \Delta + \lambda C) \cdot R = 0$ and $(K_X + \Delta + tC) \cdot R < 0$ for all $0 \leq t < \lambda$
- If the contraction induced by R is not birational, we have a Mori fiber space and we STOP.
- If the contraction induced by R is birational, we replace X by the corresponding flip/divisorial contraction and repeat the procedure.

The directed MMP

- The advantage is that for each flip $\phi : X \dashrightarrow X^+$ there is a λ such that $K_X + \Delta + \lambda C$ is nef and hence X is a log terminal model.
- Thm C_n says that there are only finitely many such models (under appropriate hypothesis) and so we expect that these flips should terminate.
- In terms of our induction, we are only allowed to assume Thm C_{n-1} and this makes the proof more technical. (An analog of special termination will be required.)

Outline of the talk

- 1 The structure of the proof
- 2 The directed MMP
- 3 Existence of models
- 4 Finiteness of models
- 5 Non-vanishing

Proof of Claim 1, Special termination

- Throughout the remainder of the talk, we will assume that Thms A_{n-1} , B_{n-1} , C_{n-1} hold so that in particular PL-flips exist in dimension n .
- Lemma.** Suppose that

$$K_X + \Delta = K_X + S + A + B$$

is DLT where A is ample and $B \geq 0$.

Then any sequence of flips $X = X_1 \dashrightarrow X_2 \dashrightarrow X_3 \cdots$ for the $K_X + \Delta$ -MMP is eventually disjoint from $S = [\Delta]$.

- Proof.** We may assume that S is irreducible and that the induced maps $S_i \dashrightarrow S_{i+1}$ are isomorphisms in codim. 1.
- We write $(K_{X_i} + \Delta_i)|_{S_i} = K_{S_i} + \Theta_i$.

Proof of Claim 1, Special termination II

- Note that as we may assume that all S_i are isomorphic in codimension 1, then Θ_i is just the strict transform of Θ_1 .
- The $K_{S_i} + \Theta_i + \lambda_i C|_{S_i}$ are weak log canonical models of $K_{S_1} + \Theta_1 + \lambda_i C|_{S_1}$ where $0 \leq \lambda \leq 1$.
- By Thm C_{n-1} there are only finitely many distinct pairs (S_i, Θ_i) .
- If Σ_i is a flipping curve for $X_i \dashrightarrow X_{i+1}$ that intersects S_i , then either $\Sigma_i \subset S_i$ or $\Sigma_i \cdot S_i > 0$ in which case the flipped curves are contained in S_{i+1} . This means that there is a center ν such that $a(\nu, S_i, \Theta_i) < a(\nu, S_{i+1}, \Theta_{i+1})$.
- Since $a(\nu, S_{i+1}, \Theta_{i+1}) \leq a(\nu, S_j, \Theta_j)$ then $S_j \not\cong S_i$ for all $j > i$. But there are only finitely many models S_j and so the flips are eventually disjoint from S_j .

Proof of Claim 1, Cont.

- **Proof of Claim 1.**
- Recall that $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$ and write $D = M + F$ where $\text{Supp}(F) \subset \text{SBs}(K_X + \Delta)$ and M is mobile.
- We may assume that X is smooth, $\Delta + D$ has simple normal crossings, $\Delta = A + B$ where A is ample and $B \geq 0$ and $\Delta \wedge M = 0$. (I.e. Δ and M have no common components.)
- For simplicity assume that M is irreducible (this avoids some linear algebra).
- Set $\Theta = \text{Supp}(M + F) \vee \Delta = S + A + B'$ where $0 \leq B' \leq B$.
 (If $G = \sum g_i G_i$, $G' = \sum g'_i G_i$ then
 $G \vee G' := \sum \max\{g_i, g'_i\} G_i$.)

Proof of Claim 1, Cont.

- we will perform a sequence of flips/contractions (for various divisors $\Theta_i \geq \Delta$) such that:
 - 1) the induced rational map $\phi : X \dashrightarrow Y$ only contracts divisors in $\text{SBs}(K_X + \Delta)$, and
 - 2) $K_Y + \Gamma$ is nef for some divisor $\Gamma = \phi_*(\Delta + \Phi)$ where $\text{Supp}(\Phi) \subset \text{SBs}(K_X + \Delta)$.
- It is then not hard to show that Y is a log terminal model for (X, Δ) .

Proof of Claim 1, Step 1

- Pick H a sufficiently ample divisor and run the $K_X + \Theta$ MMP with scaling of H .
- Recall that $\Theta = S + A + B'$ where $S = \text{Supp}(M + F)$ and $0 \leq B' \leq B$. Moreover, $\Theta = \Delta + \Psi$ with $0 \leq \Psi \leq \text{Supp}(M + F) = S$.
- Let C be a flipping curve. We have

$$0 > (K_X + \Theta) \cdot C = (D + \Psi) \cdot C$$

and so the flipping locus is contained in S .

- Therefore, this is a sequence of PL-flips that must terminate and we may assume that $K_X + \Theta$ is nef.

Proof of Claim 1, Step 2

- We wish to remove M , so we write

$$K_X + S + A + B' = K_X + F_{\text{red}} + A + B' + M_{\text{red}}$$

and we run a MMP with scaling of M_{red} .

- By a similar argument to the one above, the flipping locus is always contained in F_{red} so that this is a sequence of PL-flips that must terminate.
- Therefore, we have a model where $K_X + S + A + B' = K_X + \Delta + \Phi$ is nef and $\Phi \subset \text{SBs}(K_X + \Delta)$. As remarked above, this is a log terminal model of (X, Δ) .

Outline of the talk

- 1 The structure of the proof
- 2 The directed MMP
- 3 Existence of models
- 4 Finiteness of models**
- 5 Non-vanishing

Proof of Claims 2 and 4

- In proving Thm C (resp. C-), we need to know that for any pseudo-effective class $K_X + \Delta$ that we consider, there is an effective divisor $D \geq 0$ such that $K_X + \Delta \sim_{\mathbb{R}} D$.
- This will hold as we are assuming Thm B (resp. we are assuming that $K_X + \Delta$ is big).
- We will just show that there exists an integer $k > 0$ and rational maps $\phi_i : X \dashrightarrow Y_i$ such that if $\Delta \in \mathcal{C}$ is pseudo-effective then there is an integer i with $1 \leq i \leq k$ such that ϕ_i is a log terminal model for (X, Δ) .

Proof of Claims 2 and 4 cont.

- Since $\mathcal{C}$ is compact and the pseudo-effective cone is closed, we may work locally in a neighborhood

$$\Delta_0 \in U \subset \mathcal{C} \cap PSEF.$$

- Let $\phi : X \dashrightarrow Y$ be a log terminal model of $K_X + \Delta_0$.
- ϕ is $K_X + \Delta_0$ negative in the sense that for any exceptional divisor $E \subset X$, $a(E, X, \Delta_0) < a(E, Y, \phi_* \Delta_0)$.
- So after shrinking U , we have that ϕ is $K_X + \Delta$ negative for all $\Delta \in U$.
- So we may replace X by Y and assume that $K_X + \Delta_0$ is nef.
- In particular, as Δ_0 is big, by the base point free theorem, $K_X + \Delta_0 = f^*H$ for some morphism $f : X \rightarrow Z$ and some ample $\mathbb{R}$ -divisor H on Z .

Proof of Claims 2 and 4 cont.

- Since $\mathcal{C}$ is a rational polytope, we may assume that U is a rational polytope and so by induction on the dimension (of the affine space spanned by the polytope) there exist finitely many rational maps $\phi_i : X \dashrightarrow Y_i$ over Z such that for all $\Delta \in \partial\mathcal{C} \cap \text{PSEF}(X/Z)$ then ϕ_i is a log terminal model of (X, Δ) for some $1 \leq i \leq k$.

Proof of Claims 2 and 4 cont.

- We have seen that $K_X + \Delta_0 = f^*H$ and $K_X + \Delta_0 \sim_{\mathbb{R},Z} 0$
- Further shrinking U , we may assume that for all $\Delta \in U$ $K_X + \Delta$ is nef if and only if $K_X + \Delta$ is nef/ Z .
- Let $\Delta_0 \in \partial U$. For any $\Theta \in [\Delta_0, \Delta]$, we have

$$\Theta - \Delta_0 = (K_X + \Theta) - (K_X + \Delta_0) \sim_{\mathbb{R},Z} (K_X + \Theta)$$

so that $\Theta - \Delta_0$ is PSEF/ Z iff $K_X + \Theta$ is PSEF/ Z iff $K_X + \Delta$ is PSEF/ Z .

- If $K_X + \Theta$ is PSEF then it is PSEF/ Z and so $K_X + \Delta$ is PSEF/ Z . A log terminal model/ Z of $K_X + \Delta$ is a log terminal model/ Z of $K_X + \Theta$ and hence it is a log terminal model of $K_X + \Theta$.

Outline of the talk

- 1 The structure of the proof
- 2 The directed MMP
- 3 Existence of models
- 4 Finiteness of models
- 5 Non-vanishing**

Proof of Claim 3

- We may assume that $\Delta = A + B$ has simple normal crossings, A is ample and $B \geq 0$.
- If for any fixed $k \gg 0$ sufficiently divisible,

$$h^0(\mathcal{O}_X([mk(K_X + \Delta)] + kA))$$

is a bounded function of m , then by a result of Nakayama, we have $K_X + \Delta \equiv N_\sigma \geq 0$. Recall that $N_\sigma = N_\sigma(K_X + \Delta)$ is the limit as $\epsilon \rightarrow 0$ of divisors in $\text{SBs}(K_X + \Delta + \epsilon A)$.

- So $A' \sim_{\mathbb{R}} A + N_\sigma - (K_X + \Delta)$ is ample and $K_X + A' + B \sim_{\mathbb{R}} N_\sigma \geq 0$ and hence $K_X + A' + B$ has a log terminal model which is also a log terminal model for $K_X + A + B$.
- By the base point free theorem $K_X + A + B \sim_{\mathbb{R}} D \geq 0$.

Proof of Claim 3 cont.

- **Claim.** If $h^0(\mathcal{O}_X([mk(K_X + \Delta)] + kA))$ is not bounded, then we may assume that $\Delta = S + A + B$ where $[\Delta] = S$, A is ample, $S \not\subset N_\sigma(K_X + \Delta)$.
- **Proof.** Pick $H \sim_{\mathbb{R}} m(K_X + \Delta) + A$ with $\text{mult}_x(H) > n$ for some general $x \in X$.
- For $t \in [0, m]$ consider

$$(t+1)(K_X + \Delta) = K_X + \frac{m-t}{m}A + B + t(K_X + \Delta + \frac{1}{m}A) \sim_{\mathbb{R}}$$

$$K_X + \frac{m-t}{m}A + B + \frac{t}{m}H = K_X + \Delta_t.$$

Proof of Claim 3 cont.

- Then for some $0 < \epsilon \ll 1$ we have $K_X + \Delta_0$ is KLT;
 $\Delta_t \geq A' = (\epsilon/m)A$ for $t \in [0, m - \epsilon]$, and $K_X + \Delta_{m-\epsilon}$ has a
 log canonical center not contained in $N_\sigma(K_X + \Delta)$.
- We pass to a log resolution $\pi : Y \rightarrow X$ and write

$$K_Y + \Gamma_t = \pi^*(K_X + \Delta_t) + E_t$$

and we then cancel common components of Γ_t and $\text{SBs}(\pi^*(K_X + \Delta_t))$. The claim now follows easily.

Proof of Claim 3 cont.

- We now run a $K_X + S + A + B$ MMP with scaling of (some multiple of) A .
- For $0 < t \ll 1$ we get log terminal models $\phi_t : X \dashrightarrow Y_t$ of $K_X + S + A + B + tA$ and $S \dashrightarrow T_t$.
- S is not contained in $\text{SBs}(K_X + S + A + B)$ and so it is not contracted.
- We would like to say that for $0 < t \ll 1$, we have $Y_t \cong Y$ a log terminal model of $K_X + S + A + B$. Then by the base point free theorem, $K_X + S + A + B \sim_{\mathbb{R}} D \geq 0$.
- We can only conclude that $T_t \subset Y_t \cong T \subset Y$ in a neighborhood of T_t .
- So $K_T + \Theta = (K_{Y_t} + \Gamma_t)|_T$ is nef, Θ is big and hence $K_T + \Theta$ is semiample.

Proof of Claim 3 cont.

- We want to lift sections of $K_T + \Theta$.
- For simplicity assume $K_X + \Delta$ is $\mathbb{Q}$ -Cartier (this avoids a diophantine approximation argument).
- We may then assume that for some integer $m > 0$, we have $m(K_X + \Delta)$ is integral Weil, $m(K_Y + \Gamma)$ is Cartier and $h^0(m(K_T + \Theta)) > 0$,
- Then we look at the corresponding short exact sequence and apply Kawamata-Viehweg vanishing:

Proof of Claim 3 cont.

- We have

$$\begin{aligned} 0 \rightarrow \mathcal{O}_{Y_m}(m(K_{Y_t} + \Gamma) - T) &\rightarrow \mathcal{O}_{Y_m}(m(K_{Y_t} + \Gamma)) \\ &\rightarrow \mathcal{O}_T(m(K_T + \Theta)) \rightarrow 0. \end{aligned}$$

- Then $K_{Y_t} + \Gamma - T - (m-1)tA$ is KLT (for $0 < t \ll 1$) and $(m-1)(K_{Y_t} + \Gamma + tA)$ is nef and big.
- By Kawamata-Viehweg vanishing, we may lift sections to $m(K_{Y_t} + \Gamma)$.
- It is easy to see that these sections lift to sections of $K_X + \Delta$.